

Stock Analyzer

Abstract:

The stock market is still one of the most difficult areas to make precise predictions because of its very nature of volatility and sensitivity to a large number of internal and external variables. This project deals with the study and prediction of stock prices through deep learning methods, with special focus on Long Short-Term Memory (LSTM) networks. LSTM models are very good at extracting sequential dependencies and patterns in time series data and are thus very appropriate for stock market prediction problems.

The process consists of historical stock data collection and preprocessing, including some of the most important metrics like opening price, closing price, trading volume, and other technical indicators. The data goes through exploratory data analysis to reveal patterns and relationships. The cleaned data is then used to train an LSTM model to extract intricate temporal patterns and generate precise future price predictions.

To measure model performance, metrics like Mean Squared Error (MSE) and Mean Absolute Error (MAE) are used. The predictions of the LSTM model are compared with those of conventional models to emphasize its better performance in dealing with non-linear and non-stationary stock data. Visualizations are used to show the predicted vs. actual prices, which improves interpretability.

This project not only proves the feasibility of deep learning for financial prediction but also provides the groundwork for future inclusion of macroeconomic factors and market sentiment. The findings provide useful information for investors and analysts looking for data-driven solutions in stock market decision-making.